ALAN ANDRE POMBOZA

AI Engineer · Full-Stack Developer (.NET, Angular, Python) · MCS in Applied AI Student

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PROFESSIONAL SUMMARY

AI and full-stack engineer with 3 years of experience building production systems: election systems in .NET, Angular and Oracle 19c for Ecuador's National Electoral Council, including a computer vision anti-spoofing identity-verification module, and Databricks pipelines and AI solutions with Meta Llama models for a US client. Master of Computer Science in Applied AI student who builds and benchmarks applied ML and LLM systems. Bilingual: English (C1 certified) and Spanish (native).

TECHNICAL SKILLS

- **AI / Machine Learning:** Python, scikit-learn, embeddings, hybrid retrieval, LLM applications (Claude, OpenAI, Meta Llama, structured outputs), computer vision (liveness and spoof detection, TensorFlow.js), MLflow, model evaluation, agentic AI concepts
- **Data:** Databricks, Oracle 19c (PL/SQL, stored procedures), PostgreSQL, SQL Server, MySQL, Power BI, XtraReports
- **Back-end:** C#, .NET 8, .NET 6, .NET Framework 4.7, REST Web APIs, FastAPI, SQLAlchemy
- **Front-end:** TypeScript, JavaScript, Angular 16, React, HTML, CSS
- **DevOps & Tools:** Git, GitHub Actions, Azure DevOps (CI/CD, Repos, Boards), Docker, Postman, Scrum / Agile

PROFESSIONAL EXPERIENCE

Electoral Systems & IT Analyst (Full-Stack Developer) — **Sep 2023 – Oct 2026**
National Electoral Council (CNE) — *Quito, Ecuador*

- Built the political-organization financing accounting system: Oracle 19c stored procedures, .NET 6 / 8 Web APIs, Angular UIs.
- Developed the candidacy simulator used for all candidacies to check legal gender-parity and youth-inclusion requirements.
- Built Voto Informado, the web application where voters review candidates' work plans and proposals.
- Built the online electoral address-change system, including a machine learning and computer vision module that verifies citizen identity (person vs. spoof detection) for automated address-change requests.
- Produced executive reports in XtraReports and Power BI; deployed through Azure DevOps CI/CD pipelines and Azure Repos.

AI Software Engineer (Contract) – Client Project — **Dec 2025 – Feb 2026**
Stimulus, Inc. — *Remote, United States*

- Built Databricks data pipelines in Python and SQL and delivered production-ready AI solutions with Meta Llama models and MLflow for a US client, as one of three engineers mentored by a Senior AI/ML Engineer specializing in agentic AI.
- Translated client needs into technical solutions, from requirements and design to delivery.

Information Technology Intern — **Jan 2023 – Jun 2023**
IOET Services — *Remote*

- Built front-end and back-end features on full-stack client projects with international, distributed teams.

PROJECTS

noc-noc – NOC 2021 occupation-code finder (github.com/AlanJuker/noc-noc). Hybrid TF-IDF and embedding retrieval (reciprocal rank fusion) puts the right code in the top 5 for 93.6% of 5,468 held-out official titles and 79.4% of 4,242 real Job Bank titles, each method benchmarked against a baseline. Python, scikit-learn, embeddings, Claude API, FastAPI

signal-problems – Toronto subway delay analytics (github.com/AlanJuker/signal-problems). Resolved 99.7% of 260,000 TTC delay records (1,251 station spellings to 70 stations) without guessing; serves an API, an interactive subway map and a commute-risk check. Python, FastAPI, PostgreSQL, React, TypeScript, Docker, GitHub Actions

EDUCATION

Master of Computer Science in Applied Artificial Intelligence · *University of Niagara Falls Canada* — **2025 – Present**

Master's Degree in Software Engineering and Computer Systems · *International University of La Rioja (UNIR)* — **2023 – 2024**

Bachelor's Degree in Systems Engineering · *Pontifical Catholic University of Ecuador (PUCE)* — **2019 – 2023**

CERTIFICATIONS

AI Skills Accelerator – Awana (2026) · GitHub Foundations (2025) · Postman API Fundamentals (2024) · Python Programming – UNIR (2024) · Microsoft Azure Fundamentals AZ-900 (2023) · Scrum Foundation Professional (SFPC) – CertiProf (2023) · Application & Database Security – UNAM (2021)